Robustness survival and the Pareto frontier of causal assumptions

Session 7 — Axis Robustness

branch `experiments/survival-and-pareto` · 2026-09-10

Pre-registration	*results/axis_robustness/PREREGISTRATION.md* (§1–§9 written before any survival curve or rank correlation existed; Appendices A–F record every falsification, correction and retraction, in order).
Gate	*benchmarks.measure.fast_gate* **exclusively**, recorded in every row. *synth.runner.gate* was not used — it calls the exponential *all_valid_adjustment_sets_mpdag* internally. *fast_gate* is strictly stricter (16/46,800 disagreements, one direction), which biases **against** the hypothesis under test.
Assumption carried on every radius	*Conjecture 2 (hence Anti-Exchange Case B, verified not proved); the error is one-sided, so radii can only be too large.* All upward searches are exact iff it holds.

1. What was asked, and what the data actually supports

The brief asked for two demonstrations: that `r_val` ranks average-case structural survival where SHD and |K| fail, and that precision trades off against robustness along a Pareto frontier ("the fragility of precision").

The first is supported. The second is false, and the reason it is false is the more interesting result of the two.

Prediction	Verdict	Evidence
P1 r_val positively ranks survival	**SUPPORTED**	$\tau_b > 0$ in 9/9 strata; 8/9 under the conservative endpoint
P2 r_val outranks |K|	**SUPPORTED**	9/9 strata; 7/9 under the conservative endpoint
P3 |K| is a null predictor (project hypothesis H4)	**NOT SUPPORTED**	CI covers 0 in only 3/9; |K| is *negatively* associated
P4 tiered AUC variance > flip	**NOT COMPARABLE**	unit mismatch; no verdict forced (Appendix E.4)
P5 claim-axis and hop-axis agree in sign	**PASS**	6/6 strata, independent subsample
P6 a non-trivial efficiency/robustness frontier exists	**FALSIFIED**	utility exactly invariant in 32/32 strata
P6′ frontier exists for *achievable* efficiency	**FALSIFIED**	invariant again; Phase 2 abandoned per pre-registered trigger
P7 / P7′ aggressive BK trades utility for radius	**UNTESTABLE**	one axis is constant; census reported instead

2. Phase 1 — survival under corrupted background knowledge

2.1 Design

An analyst holds CPDAG $\hat{C}$, asserts knowledge K, Meek-closes to G_0, and reads Z* = optimal_adjustment_set_mpdag(G_0, x, y) **once, at depth 0, then holds it fixed**. K is then corrupted and we ask whether the set they already committed to survives — is_gac_valid_mpdag(G, x, y, Z*), polynomial, no enumeration of [G].

1,978 instances; 3,136,000 sampled states; 200 reps per depth; 1,656 s. Instances span component size 6–12, three separations, coverage ∈ {0.5, 1.0}, and a **base wrongness** b ∈ {0, 0.10, 0.25} applied before the sweep so that SHD(G_0, truth) is not constant — at coverage 1.0 with truthful K, G_0 *is* the ground-truth DAG and SHD ≡ 0, which would have made the baseline comparison rigged by construction.

Two arms, never merged: **flip** (uniform, independent, targeted depth) and **tiered** (block-correlated, re-binned on corruption_rate; see §2.5).

2.2 P1 — r_val ranks survival

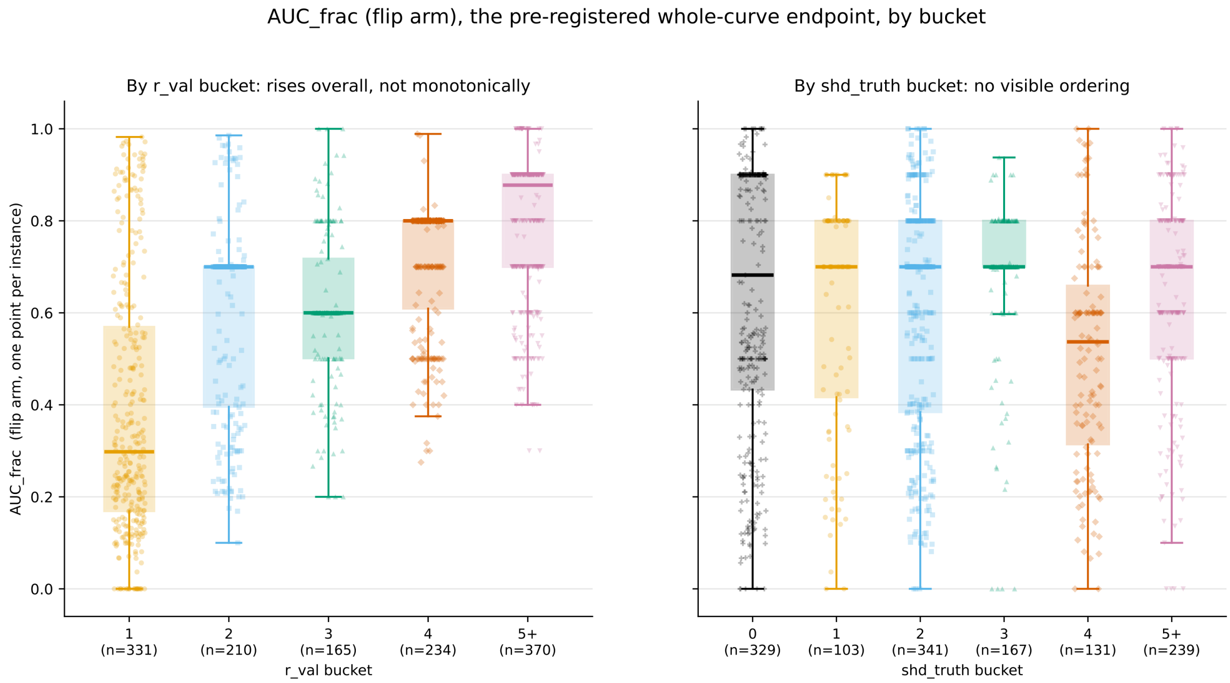
τ_b(AUC, r_val), 10,000-resample bootstrap over instances, seed 0:

stratum	τ_b	95% CI	n
flip cov=0.5 bw=0.00	+0.714	[+0.625, +0.798]	240
flip cov=1.0 bw=0.00	+0.519	[+0.442, +0.591]	240
flip cov=1.0 bw=0.10	+0.388	[+0.296, +0.478]	236
flip cov=0.5 bw=0.10	+0.380	[+0.265, +0.488]	240
flip cov=1.0 bw=0.25	+0.346	[+0.209, +0.472]	134
flip cov=0.5 bw=0.25	+0.031	[−0.114, +0.169]	220
tiered n_tiers=3	+0.811	[+0.787, +0.833]	238
tiered n_tiers=2	+0.799	[+0.761, +0.834]	190
tiered n_tiers=4	+0.777	[+0.753, +0.798]	240

Radius assumption: Conjecture 2 (hence Anti-Exchange Case B, verified not proved); the error is one-sided, so radii can only be too large.

Positive in 9/9; CI excludes zero in 8/9. The exception is the highest-corruption, lowest-coverage flip cell, where the association is indistinguishable from zero — reported, not dropped.

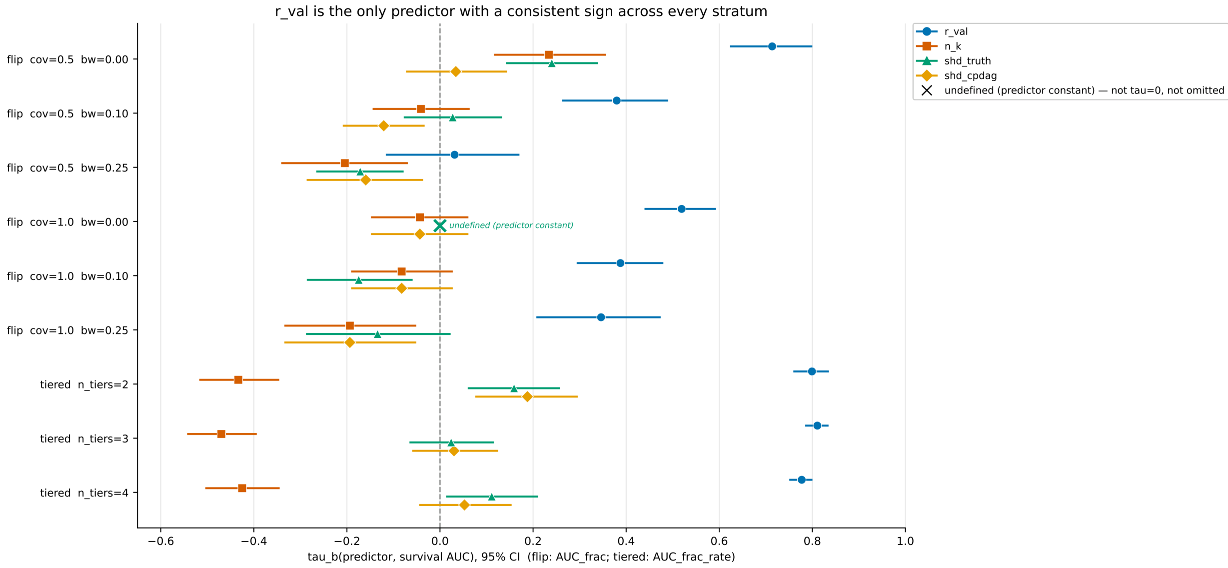
Effect sizes move monotonically: for tiered n_tiers=3, median AUC rises 0.67 → 0.94 → 0.99 → 0.98 → 1.00 across r_val buckets 1→5+.



Flip arm; AUC_frac per instance; boxes = quartiles/median, n per tick; points jittered (seed 0). shd_truth = 0 also includes every coverage=1.0/base-wrongness=0 case, where it is ≡ 0 by construction — a separate reason it cannot rank, not merely tangling. r_val carries Conjecture 2 (hence Anti-Exchange Case B, verified not proved); the error is one-sided, so radii can only be too large. F3's per-stratum tau_b remains primary evidence for P1/P2.

Figure 1. AUC_frac by r_val bucket vs. by shd_truth bucket — the endpoint. The r_val panel (left) rises overall but **not monotonically** — the median dips at bucket 3 — while the shd_truth panel shows no comparable structure. Conjecture 2 (hence Anti-Exchange Case B, verified not proved); the error is one-sided, so radii can only be too large.

The primary stratified evidence for P1/P2. Figure 2 below breaks the same association down by predictor across all nine strata — this is the figure the verdicts above rest on, not the pooled scatter above it.



analysis_tau_primary.csv, is_primary rows only, 10,000-resample bootstrap. r_val carries Conjecture 2 (hence Anti-Exchange Case B, verified not proved); the error is one-sided, so radii can only be too large.

Figure 2. τ_b per stratum, four predictors: r_val, |K|, shd_truth = SHD(G_0, true DAG), and shd_cpdag = SHD($\hat{C}$, G_0), the orientations committed including Meek propagation. 95% bootstrap CIs over instances. Endpoint is AUC_frac for flip strata and AUC_frac_rate for tiered strata, as the axis states. This is the primary evidence figure for P1 and P2: r_val's interval is positive in 8 of 9 strata and dominates |K| in every stratum shown. Conjecture 2 (hence Anti-Exchange Case B, verified not proved); the error is one-sided, so radii can only be too large.

2.3 P2/P3 — the baselines fail, but not the way the brief assumed

r_val outranks |K| in **9/9** strata. But |K| is **not inert**, so H4's framing is wrong: its association with survival is real and **negative** — flip −0.194/−0.205 at high corruption, tiered −0.425 to −0.470, all with CIs excluding zero.

> More asserted claims means *worse* survival, not better.

SHD(G_0, truth) fails for a different reason again: it is `undefined (predictor constant)` at b = 0 **by construction**, and elsewhere its sign flips across strata (+0.240, +0.027, −0.175, −0.171, −0.134). It has no consistent direction. That is a fairer indictment than “SHD tangles”: a third of the design cannot score it at all.

2.4 The discordance case

`c06_s01_cov050_seed00000007_bw000`. shd_truth = 0 — G_0 *is* the ground-truth DAG, the analyst's knowledge is perfectly correct — with n_k = 2. Yet AUC_frac = 0.0: S(1) = 0.0 on 107 non-contradictory samples, S(2) = 0.0 on 200. Every corruption invalidates the committed set, and r_val = 1 says exactly that.

> **Zero SHD is compatible with maximal fragility.** SHD and |K| describe how much was asserted; r_val describes how much can go wrong before the answer breaks.

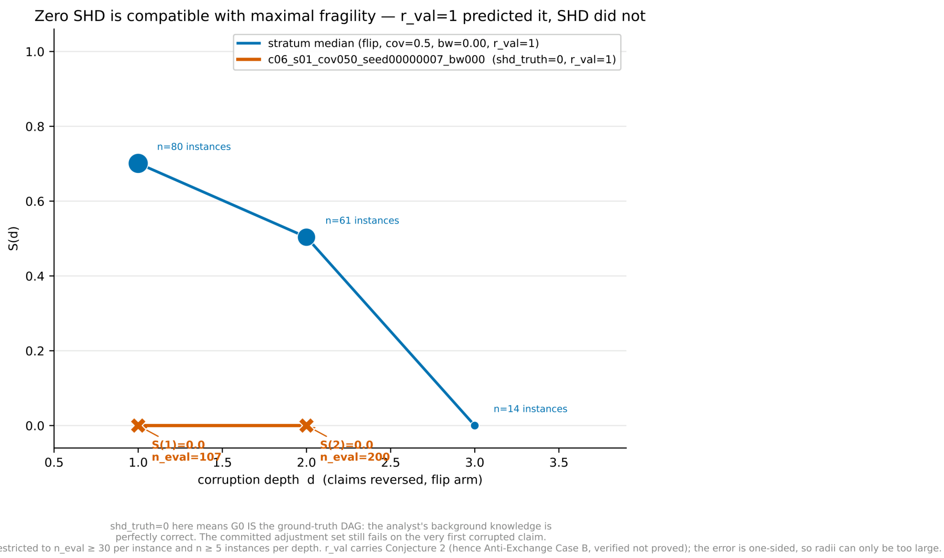


Figure 3. The verified discordance case — zero structural error, zero survival. Conjecture 2 (hence Anti-Exchange Case B, verified not proved); the error is one-sided, so radii can only be too large.

> A second, opposite spotlight (AUC_frac = 1.0 at r_val = 8) was **withdrawn on verification**: its AUC rested on depths with n_eval of 35, 10, 1 and 1. Four surviving draws are not a survival curve.

2.5 Two defects found and corrected mid-analysis

The tiered depth metric was broken (Appendix E). It counted assertions added or reversed but not *removed*; tiered relocates nodes, and relocation into a shared tier silently deletes a cross-tier assertion. Its d = 0 bin was contaminated by every corruption rate and contained genuine failures. Re-binning on corruption_rate — the knob actually turned — gives a clean design: contradiction rises monotonically 0.000 → 0.852, S falls 1.000 → 0.618, S = 1.000 exactly at rate 0 for all 668 instances individually, n = 133,600 per rate. Tiered τ values **survive** the correction (0.777–0.811). The flip arm was never affected.

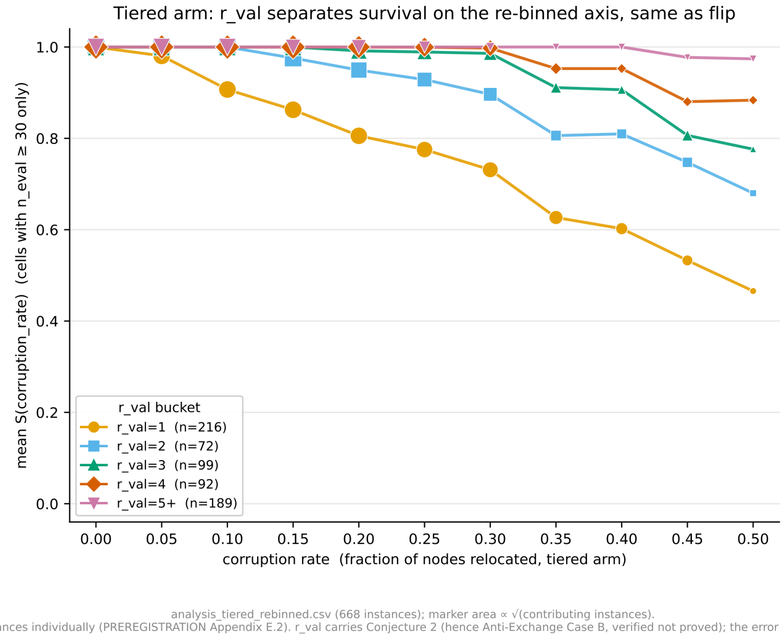


Figure 4. The tiered arm on its own rate axis, after re-binning on corruption_rate rather than the defective node-count depth. Conjecture 2 (hence Anti-Exchange Case B, verified not proved); the error is one-sided, so radii can only be too large.

Effective sample size varies with r_val (Appendix F). τ(r_val, usable depths) runs from +0.318 to −0.438 depending on stratum, so the bias runs *against* the hypothesis in all coverage-1.0 and tiered strata and *for* it in the two strata carrying the largest τ. Under the pre-specified control AUC_frac_usable (depths with n_eval ≥ 30), **P1 falls 9/9 → 8/9 and P2 9/9 → 7/9**. Weakened, not overturned. Per Appendix F.3 the conservative figure is the one quoted in any summary claim.

2.6 P5 — the units question

r_val counts BFS hops; corruption counts claims. These are not the same axis, and radius_local_up_fast cannot measure the distance because it searches *upward* only while a flip-corrupted state is not a refinement of G_0. Exact hop distance therefore required full space enumeration on a small-component subsample (1,256 instances, component size 3–7; spaces of 32–607 elements, 0.007–12.2 s to build).

One corrupted claim ≈ **2.9 hops** (sd 1.01), converging to exactly $2 \cdot d$ at $d \geq 5$. **P5 passes in 6/6 strata**, signs agreeing, τ positive on both axes. Cross-checked against the main dataset: 6/6 sign agreement, 4/6 rough magnitude agreement; the two divergent cells are plausibly a component-size effect (main spans 6–12, subsample 3–7) — reported, not resolved.

Radius assumption: Conjecture 2 (hence Anti-Exchange Case B, verified not proved); the error is one-sided, so radii can only be too large.

2.7 The finding that was not predicted

Contradiction rates by corruption depth, **flip arm only**:

depth	1	2	3	5	8	12
rate	0.627	0.777	0.876	0.892	0.983	1.000

At d = 1 and base wrongness 0.00/0.10/0.25 the rates are 0.653/0.647/0.563.

*(Corrected 2026-09-10. This table originally printed 0.491/0.773/0.896/0.930/0.986/1.000 under a “flip arm” label. Those are the **both-arms pooled** figures, and the pool is computed over the tiered arm's defective d column — the exact cross-arm merge forbidden everywhere else in this report. Caught while building the session figures, which recompute every plotted number from source. Flip-only figures are shown above; the base-wrongness breakdown was already flip-only and is unchanged. See Appendix G.)*

Most orientation errors are self-revealing. Reversing a single truthful claim renders the knowledge set inconsistent with the CPDAG about 6 times in 10. Meek closure fails and the analyst finds out.

And among corrupted sets that *stay* consistent, median survival is **0.952** — most survivors are harmless. The picture is a two-stage filter: most errors announce themselves, most of the rest are benign, and the dangerous case is **consistent *and* invalidating**. That is precisely the case r_val is defined over, since its space contains only consistent MPDAGs.

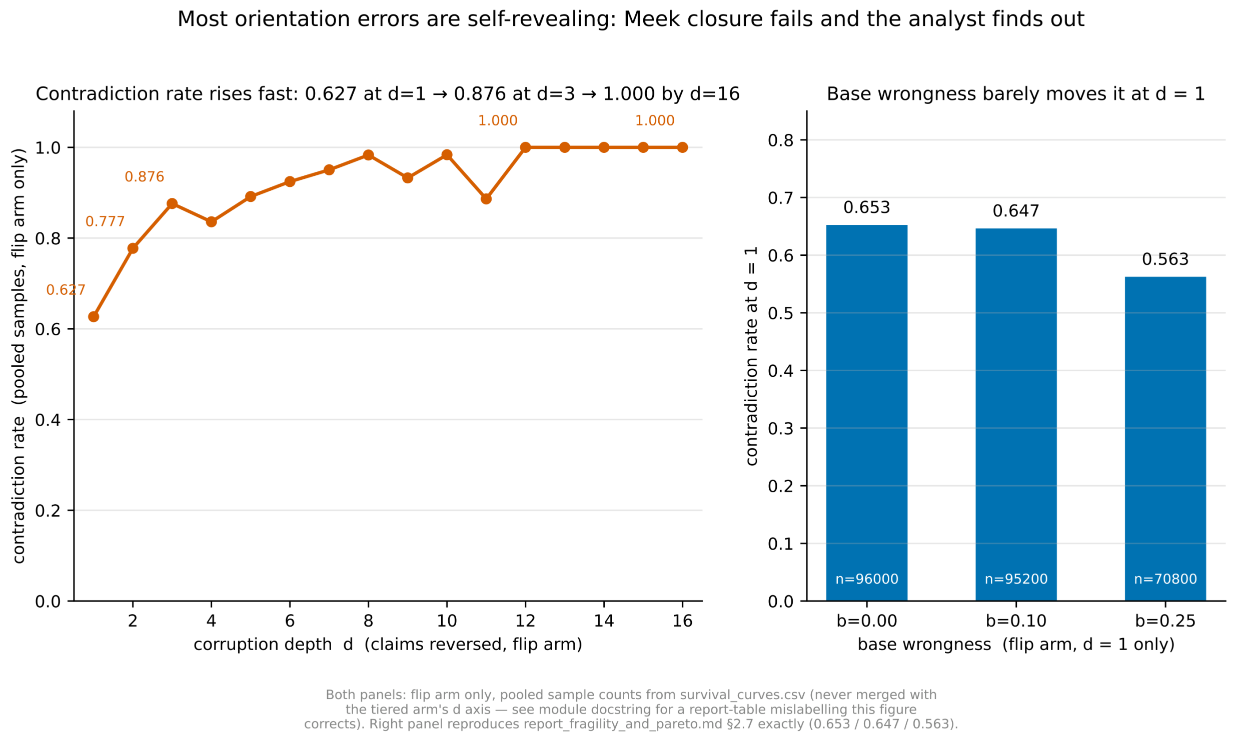


Figure 5. Contradiction rate by depth, flip arm only, after the pooling correction above.

This also dissolves an apparent tension. r_val = 1 is common while average survival at d = 1 is high. Both are true: r_val is a **worst-case** distance to the nearest failing graph, S(d) an **average-case** probability over a shell. A failure at distance 1 says nothing about how much of the shell fails. That the worst case predicts the average case is the finding, not a tautology.

3. Phase 2 — there is no Pareto frontier

3.1 Falsified twice

Round 1 (24 base CPDAGs, 1,087 truthful proposals, 17,680 SEM draws): strata with more than one distinct utility_median: **0 of 24**, while r_val varied in 18. Round 2 redefined utility as *achievable* efficiency over a capped 12-set menu: **0 of 32** strata varied in achievable utility, in the count of valid menu members, or in the winning member — while r_val varied in 23. The pre-registration's trigger was explicit, so Phase 2 was **abandoned rather than redefined a third time**.

3.2 Why — and this is the substantive result

Every truthful proposal leaves the true DAG inside $[G_0]$. optimal_adjustment_set_mpdag returns non-None exactly when all extensions agree; if they agree, the common answer *is* the true DAG's optimal set. Every identifying truthful proposal therefore yields the same Z, hence the same asymptotic variance.

For truthful background knowledge, statistical efficiency is an invariant of ($\hat{C}$, x, y) — not a function of what the analyst asserts. Knowledge moves the robustness axis and only the robustness axis.

So the trade-off is not precision against robustness. Precision is *fixed*. What knowledge purchases is **identifiability** — 385 of 1,431 proposals (26.9%) could justify no valid set at all. The operative trade-off is **identifiability against robustness, and it is a step, not a frontier**: assert enough to identify, and every further claim is pure r_val cost at zero precision gain.

This converges with Phase 1's P3 from the opposite direction: there, more claims measurably *reduced* survival. Two independent designs, same conclusion — marginal truthful background knowledge is not free, and its price is paid entirely in robustness.

Testing the step hypothesis properly — does r_val fall monotonically with proposal size beyond the minimum identifying set? — is a separate question and is **not claimed here**.

3.3 P7′ — the census stands even though the contrast cannot be run

Round 1 found **zero** “aggressive” proposals, an artifact of component_generator, which builds the undirected component purely from X's upstream confounders and keeps X→Y and its descendants outside it by construction.

On real structure the picture is different. Sampling ≤200 descendant-ordered pairs per network across the 39-network corpus (M-bias excluded as an ADMG), **28 networks admit at least one (x, y) pair with an undirected CPDAG edge on a directed causal path — 396 pairs**, densest in paths (96), Polzer_2012 (64), Kampen_2014 (33), child (22), sachs (22), insurance (21). Eight real-network bases were gated and yielded 215 aggressive and 129 conservative proposals, so the empty cell is genuinely filled — but utility is invariant there too (0 of 8 strata vary), so the intended contrast cannot be run. The residual r_val-only contrast is non-directional (2 bases lower for aggressive, 2 tied, 1 reversed) and is **inconclusive**.

Radius assumption: Conjecture 2 (hence Anti-Exchange Case B, verified not proved); the error is one-sided, so radii can only be too large.

4. What is not claimed

- **No cross-arm detectability comparison.** An earlier draft claimed correlated errors are harder to detect than uniform ones (0.354 vs 0.653 at d = 1). The tiered half came from the contaminated bin, and the comparison was never sound in principle: one arm relocates a fraction of *nodes*, the other reverses a count of *claims*. Retracted in Appendix E.3. Establishing it needs a corruption process parameterised identically across both arms — a design change, not a reanalysis.
- **No claim of perfect ranking.** The endpoint is a rank correlation with a bootstrap interval. One stratum is null.
- **Synthetic structure only** for Phase 1. Real networks appear only in the P7′ census.
- **Oracle CI throughout.** No finite-sample discovery; causal sufficiency assumed. These remain the project's largest scope limits and this work does not touch them.
- **AUC_frac is not quoted alone.** The conservative AUC_frac_usable figure governs summary claims (Appendix F.3).

5. Provenance

All results under `results/axis_robustness/`: `survival_instances.csv` (1,978), `survival_curves.csv` (14,066), `survival_samples.csv.gz` (3,136,000 rows, 48 MB compressed), `hops_samples.csv.gz` (6 MB), `analysis_tau_primary.csv`, `analysis_tau_secondary.csv`, `analysis_tiered_rebinned.csv` (668), `analysis_decomposition.csv`, `analysis_discordance.csv`, `analysis_effect_sizes.csv`, `pareto_proposals.csv` (1,087), `pareto_sems.csv` (17,680), `pareto2_proposals.csv` (1,431), `pareto2_menus.csv` (243), `pareto2_aggressive_census.csv` (39), `hops_p5.csv`, `hops_instances.csv`, `hops_cost_census.csv`, plus manifests capturing git SHA, interpreter, platform, library versions and RADIUS_CONVENTION.

`num_workers: 1`, `random_seed: 0`, CP-SAT single-threaded, no global RNG anywhere in the live tree. Determinism verified across `PYTHONHASHSEED 0 / 12345` for every stage; bootstrap CIs bit-identical across independent runs. The decomposition identity S_contra_as_fail(d) = (1 − contradiction_rate(d)) · S(d) holds to 2.2e-16 on all 8,841 rows where S is defined.

Known incident. The first survival sweep was destroyed when its worker relaunched and overwrote its own outputs in place; it was regenerated (run 2) and reproduces the original's pooled statistics exactly. The relaunch also fixed a real defect — run 1's instance_id was not unique — so several Appendix C statistics computed by grouping on it were wrong and are corrected in Appendix D, including a retracted endpoint swap. Full account in Appendices C–D.

Environment

Python 3.9.6, reportlab 5.0.0. Every table above is transcribed verbatim from `report_fragility_and_pareto.md` (committed at `ed9270d`) rather than re-derived, for the reasons given in this module's docstring; no CSV, the markdown, or any figure was modified to build this document.

Appendix — supplementary figure

Not referenced by any numbered section above; included for completeness as a per-depth breakdown underlying the whole-curve endpoint used throughout §2.

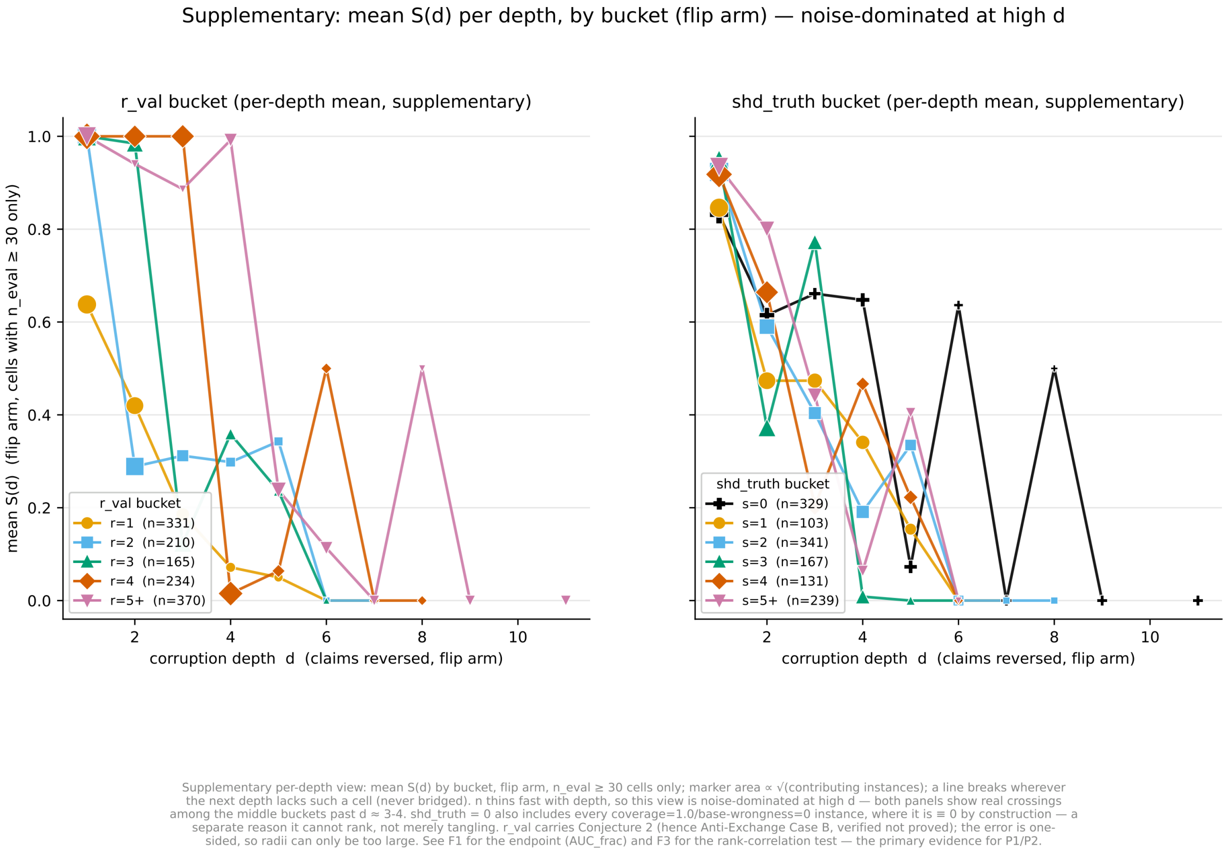


Figure 6 (supplementary). Survival curves S(d) by r_val bucket, per corruption depth — the per-depth view that Figure 1's whole-curve AUC_frac endpoint summarises. Supplementary: see §2.2 for the primary evidence. Conjecture 2 (hence Anti-Exchange Case B, verified not proved); the error is one-sided, so radii can only be too large.

Commits in this series

commit	date	subject
ed9270d	2026-09-10	Session 7 figures, and fix a mislabeled table in the report
e95f6af	2026-09-10	Axis Robustness: survival curves, and the Pareto premise falsified